Superoxide Dismutase Assay Kit

Item No. 706002

www.caymanchem.com
Customer Service 800.364.9897
Technical Support 888.526.5351
1180 E. Ellsworth Rd · Ann Arbor, MI · USA

TABLE OF CONTENTS

GENERAL INFORMATION
3 Materials Supplied
4 Safety Data
4 Precautions
4 If You Have Problems
5 Storage and Stability
5 Materials Needed but Not Supplied

INTRODUCTION
6 Background
7 About This Assay

PRE-ASSAY PREPARATION
8 Reagent Preparation
10 Sample Preparation
14 Sample Matrix Properties

ASSAY PROTOCOL
15 Plate Set Up
17 Standard Preparation
18 Performing the Assay

ANALYSIS
20 Calculations
21 Performance Characteristics

RESOURCES
22 Interferences
23 Troubleshooting
24 References
25 Plate Template
26 Notes
27 Warranty and Limitation of Remedy

GENERAL INFORMATION

Materials Supplied

Item Number	Item	96 Well Quantity/Size	480 Well Quantity/Size	Storage Temperature
706001	SOD Assay Buffer (10X)	1 vial/5 ml	1 vial/20 ml	-20°C
706003	SOD Assay Sample Buffer (10X)	1 vial/5 ml	1 vial/10 ml	-20°C
706004	SOD Assay Radical Detector	1 vial/250 µl	2 vials/250 µl	-20°C
706005	SOD Standard	1 vial/100 µl	2 vials/100 µl	-20°C
706006	SOD Assay Xanthine Oxidase	3 vials/50 µl	6 vials/50 µl	-20°C
400014	96-Well Solid Plate (Colorimetric Assay)	1 plate	5 plates	RT
400012	96-Well Cover Sheet	1 ea	5 ea	RT

If any of the items listed above are damaged or missing, please contact our Customer Service department at (800) 364-9897 or (734) 971-3335. We cannot accept any returns without prior authorization.

WARNING: THIS PRODUCT IS FOR RESEARCH ONLY - NOT FOR HUMAN OR VETERINARY DIAGNOSTIC OR THERAPEUTIC USE.

Safety Data

This material should be considered hazardous until further information becomes available. Do not ingest, inhale, get in eyes, on skin, or on clothing. Wash thoroughly after handling. Before use, the user must review the complete Safety Data Sheet, which has been sent *via* email to your institution.

Precautions

Please read these instructions carefully before beginning this assay.

The reagents in this kit have been tested and formulated to work exclusively with Cayman Chemical's Superoxide Dismutase Assay Kit. This kit may not perform as described if any reagent or procedure is replaced or modified.

If You Have Problems

Technical Service Contact Information

Phone: 888-526-5351 (USA and Canada only) or 734-975-3888

Email: techserv@caymanchem.com

In order for our staff to assist you quickly and efficiently, please be ready to supply the lot number of the kit (found on the outside of the box).

Storage and Stability

This kit will perform as specified if stored as directed in the **Materials Supplied** section (see page 3) and used before the expiration date indicated on the outside of the box.

Materials Needed But Not Supplied

1. A plate reader capable of measuring an absorbance at 450 nm
2. Adjustable pipettes; multichannel or repeating pipettor recommended
3. A source of pure water; glass-distilled water or HPLC-grade water is acceptable. *NOTE: UltraPure Water is available for purchase from Cayman (Item No. 400000).*
4. An orbital microplate shaker
5. Materials used for **Sample Preparation** (see page 10)

INTRODUCTION

Background

Superoxide dismutases (SODs) are metalloenzymes that catalyze the dismutation of the superoxide anion to molecular oxygen and hydrogen peroxide and thus form a crucial part of the cellular antioxidant defense mechanism.[1] Three types of SODs have been characterized according to their metal content: copper/zinc (Cu/Zn), manganese (Mn), and iron (Fe).[2,3] SOD is widely distributed in both plants and animals.[1] In humans, there are three isoforms of SOD: cytosolic Cu/Zn SOD, also known as SOD1, mitochondrial Mn-SOD, also known as SOD2, and extracellular SOD (EC-SOD), also known as SOD3.[2,3] Cu/Zn SOD and Mn-SOD are ubiquitously expressed and EC-SOD is found in the interstitial spaces of tissues and extracellular fluids, accounting for the majority of the SOD activity in plasma, lymph, and synovial fluid.[3] The presence of sufficient amounts and high turnover of SOD in cells and tissues typically keeps the concentration of superoxide very low.[4] However, when nitric oxide (NO) concentrations are high or SOD levels are low or absent, NO reacts with superoxide to form the oxidizing and nitrating agent peroxynitrite, which nitrates tyrosine residues in proteins to form nitrotyrosine. Mutations in *SOD1* are associated with amyotrophic lateral sclerosis (ALS) and single nucleotide polymorphisms in *SOD2* have been found in patients with non-Hodgkin's lymphoma, lung cancer, or colorectal cancer.[5-7]

About This Assay

Cayman's Superoxide Dismutase Assay Kit provides a convenient method of measuring SOD activity in plasma, serum, erythrocyte lysates, tissue homogenates, and cell lysates. The assay relies on xanthine oxidase to produce superoxide radicals, which then reduce a tetrazolium salt to a formazan dye, detectable at 450 nm (see Figure 1). SOD in the sample inhibits this reduction by catalyzing the dismutation of superoxide radicals. Thus, the degree of inhibition is directly proportional to SOD activity. The SOD Assay Kit measures all three types of SOD (Cu/Zn-, Mn-, and Fe-SOD). Mitochondrial Mn-SOD can be assayed separately following the procedure outlined under **Sample Preparation** (see page 10).

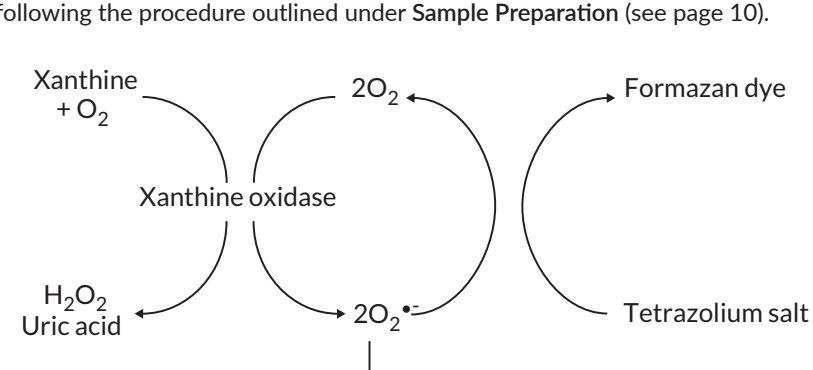


Figure 1. Assay scheme

PRE-ASSAY PREPARATION

Reagent Preparation

NOTE: It is normal for the concentrated buffers to contain crystalline salts after thawing. These will completely dissolve upon dilution with pure water.

1. **SOD Assay Buffer (10X) - (Item No. 706001)**

   This vial contains concentrated SOD Assay Buffer. Thaw the SOD Assay Buffer (10X) at room temperature and mix 5 ml with 20 ml of pure water. This diluted assay buffer, Assay Buffer (2X), should be used for diluting the radical detector. The diluted buffer will be stable for two months when stored at 4°C.

2. **SOD Assay Sample Buffer (10X) - (Item No. 706003)**

   This vial contains concentrated SOD Assay Sample Buffer. Thaw the SOD Assay Sample Buffer (10X) at room temperature and mix 5 ml with 45 ml of pure water. The diluted sample buffer, Sample Buffer (1X), should be used in the assay and the preparation of samples, SOD standards, and xanthine oxidase. This diluted buffer will be stable for six months when stored at 4°C.

3. **SOD Assay Radical Detector - (Item No. 706004)**

   Each vial contain 250 µl of a tetrazolium salt solution. If not using all at once, prepare aliquots and store at -20°C. Limit freeze-thaw cycles to three.

   Prior to use, prepare the radical detector mix by adding 50 µl of the SOD Assay Radical Detector to 9.95 ml of Assay Buffer (2X) (see Table 1, page 10). This is a sufficient volume to assay a full plate. Scale as needed. The radical detector mix will be stable for two hours when stored at room temperature and protected from light.

4. **SOD Standard - (Item No. 706005)**

   Each vial contains 100 µl of 575 U/ml bovine erythrocyte Cu/Zn SOD. Once thawed, it is ready to use for preparation of the standard curve as described in **Standard Preparation** section on page 17. If not using the all at once, prepare aliquots and store at -20°C. Limit freeze-thaw cycles to two.

5. **SOD Assay Xanthine Oxidase - (Item No. 706006)**

   Each vial contains 50 µl of SOD Assay Xanthine Oxidase. Prior to use in the assay, add 1.15 ml of Sample Buffer (1X) to the vial and pipette gently to mix. Then, further dilute 10-fold by mixing 1.2 ml with 10.8 ml of Sample Buffer (1X). *Do not vortex*. This is a sufficient volume to assay a full plate. Place the diluted enzyme on ice where it will be stable for one hour. Discard any remaining thawed/diluted xanthine oxidase.

6. **Radical Detector Mix**

   NOTE: The addition of potassium cyanide to a final concentration of 1-3 mM in the assay will inhibit both Cu/Zn SOD and EC-SOD resulting in the detection of only Mn-SOD activity.[3,8]

   Prepare Radical Detector Mix according to Table 1, below. Scale as needed

7. **Radical Detector Mix + Inhibitor (Optional)**

   This mix should only be prepared if including an inhibitor in the assay. First, prepare an inhibitor stock solution at a concentration 115X the final concentration. Then, prepare the radical detector + inhibitor mix following Table 1, below. Use this mix for all inhibitor-treated samples only. Do not use for standards. Scale as needed.

Reagent	Radical Detector Mix	Radical Detector + Inhibitor Mix (optional)
Radical Detector	50 µl	10 µl
Assay Buffer (2X)	9,950 µl	1,950 µl
115X Inhibitor	--	40 µl
Final Volume	10,000 µl	2,000 µl

Table 1. **Preparation of the radical detector mixes**

Sample Preparation

It is recommended to assay samples *in the absence of Xanthine Oxidase* to generate a sample background.

Tissue Homogenate

Total SOD measurement (cytosolic and mitochondrial)

1. Prior to dissection, rinse the tissue with PBS, pH 7.4, to remove any red blood cells and clots.
2. Homogenize the tissue in 5-10 ml of cold 20 mM HEPES buffer, pH 7.2, containing 1 mM EGTA, 210 mM mannitol, and 70 mM sucrose per gram of tissue. Alternatively, tissues may be snap-frozen in liquid nitrogen and homogenized using a Precellys® homogenizer following the manufacturer's protocol.
3. Centrifuge at 1,500 x g for five minutes at 4°C.
4. Remove the supernatant for assaying and store on ice. This supernatant will contain total SOD activity (cytosolic and mitochondrial). To measure individually, proceed to step 5. If not assaying on the same day, freeze the sample at -80°C. The sample will be stable for at least one month.

Cytosolic SOD measurement

5. Centrifuge the supernatant from step 4 at 10,000 x g for 15 minutes at 4°C. The resulting supernatant will contain cytosolic SOD, and the pellet will contain mitochondrial SOD.[9] Remove the supernatant for use in the assay. If not assaying the same day, freeze the sample at -80°C. The samples will be stable for one month.

   NOTE: *Supernatants from step 4 and 5 will require dilution with Sample Buffer 1X to fall within range of the standard curve. It is recommended to assay several dilutions of each sample with a minimum dilution of 1:20. Samples will vary.*

Mitochondrial SOD measurent

6. To assay mitochondrial SOD measurement, homogenize the mitochondrial pellet from step 5 in a small volume cold buffer (*i.e.*, 20 mM HEPES, pH 7.2, containing 1 mM EGTA, 210 mM mannitol, and 70 mM sucrose). If not assaying the same day, freeze the sample at -80°C. The samples will be stable for one month.

   NOTE: Suspension may require dilution with Sample Buffer 1X to fall within range of the standard curve. It is recommended to assay several dilutions using Sample Buffer (1X) of each sample with a minimum dilution of 1:10. Samples will vary.

Cell Lysate

Total SOD measurement (cytosolic and mitochondrial)

1. Collect cells (~5 x 10^6) by centrifugation (*i.e.*, 1,000-2,000 x g for 10 minutes at 4°C). For adherent cells, use a cell scraper instead of proteolytic enzymes.
2. Homogenize or sonicate the cell pellet in 1 mlcold 20 mM HEPES buffer, pH 7.2, containing 1 mM EGTA, 210 mM mannitol, and 70 mM sucrose.
3. Centrifuge at 1,500 x g for five minutes at 4°C.
4. Remove the supernatant for assaying and store on ice. If not assaying on the same day, freeze the sample at -80°C. The sample will be stable for at least one month.

Cytosolic SOD measurement

5. Centrifuge the supernatant from step 4 at 10, 000 x g for 15 minutes at 4°C. The resulting supernatant will contain cytosolic SOD, and the pellet will contain mitochondrial SOD.[9] Remove the supernatant for use in the assay. If not assaying the same day, freeze the sample at -80°C. The samples will be stable for one month.

   NOTE: Supernatants from steps 4 and 5 may require dilution with Sample Buffer 1X to fall within range of the standard curve. It is recommended to assay several dilutions using Sample Buffer 1X of each sample with a minimum dilution of 1:2. Samples will vary.

Mitochondrial SOD measurent

6. To assay mitochondrial SOD measurement, homogenize the mitochondrial pellet from step 5 in a small volume of cold buffer (*i.e.*, 20 mM HEPES, pH 7.2, containing 1 mM EGTA, 210 mM mannitol, and 70 mM sucrose). If not assaying the same day, freeze the sample at -80°C. The samples will be stable for one month.

   NOTE: Suspension may require dilution with Sample Buffer 1X to fall within range of the standard curve. It is recommended to assay several dilutions of each sample with a minimum dilution of 1:2. Samples will vary.

Plasma and Erythrocyte Lysate

1. Collect blood using an anticoagulant, such as heparin, EDTA, or citrate.
2. Centrifuge the blood at 700-1,000 x g for 10 minutes at 4°C. Pipette off the top yellow plasma layer without disturbing the white buffy layer. Store plasma on ice. If not assaying the same day, store at -80°C. Avoid repeated freeze/thaw cycles. The plasma sample will be stable for at least one month.

   NOTE: Plasma should be diluted 1:5 with Sample Buffer (1X) before assaying for SOD activity.
3. Remove the white buffy layer (leukocytes) and discard.
4. Lyse the erythrocytes (red blood cells) in four times its volume of ice-cold pure water.
5. Centrifuge at 10,000 x g for 15 minutes at 4°C.
6. Collect the supernatant (erythrocyte lysate) for assaying and store on ice. If not assaying the same day, freeze at -80°C. The sample will be stable for at least one month.

   NOTE: The erythrocyte lysate should be diluted 1:100 with Sample Buffer (1X) before assaying for SOD activity. Assaying several dilutions is recommended.

Serum

1. Collect blood without using an anticoagulant.
2. Allow blood to clot for 30 minutes at 25°C.
3. Centrifuge the blood at 2,000 x g for 15 minutes at 4°C. Pipette off the top yellow serum layer without disturbing the white buffy layer. Store serum on ice. If not assaying the same day, store at -80°C. Avoid repeated freeze/thaw cycles. The sample will be stable for at least one month.

   NOTE: Serum should be diluted 1:5 with Sample Buffer (1X) before assaying for SOD activity. Assaying several dilutions is recommended.

Sample Matrix Properties

Parallelism

To assess parallelism, mouse brain, heart, and kidney and human erythrocytes, plasma, and serum were assayed at multiple dilutions using the SOD Assay Kit. All samples were prepared as in **Sample Preparation** (pages 10-13). Concentrations were plotted as function of sample dilution. The results are shown below.

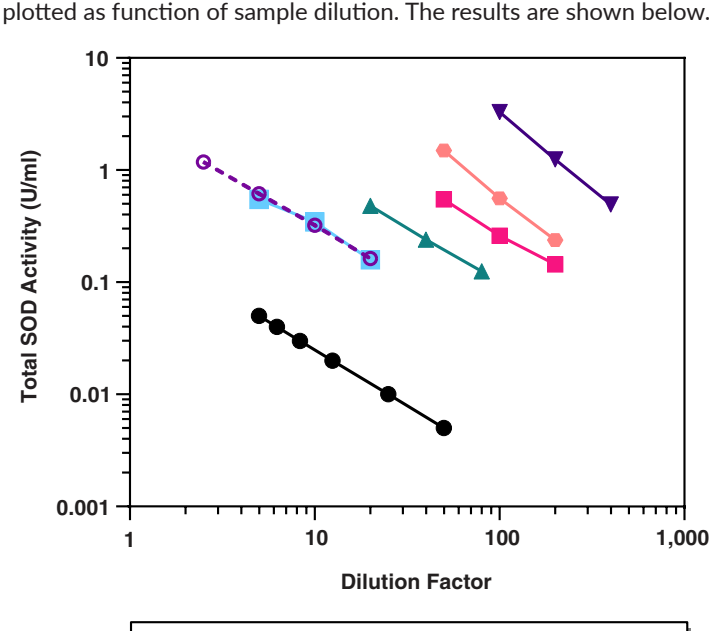


Figure 2. Parallelism of various matrices in the SOD Assay Kit

ASSAY PROTOCOL

Plate Set Up

There is no specific pattern for using the wells on the plate. It is suggested that each sample and standard be assayed in duplicate and that the contents of each well are recorded on the template sheet provided on page 25. A typical layout of samples to be measured in duplicate is provided below. It is recommended to run a Sample Background to eliminate absorbance at 450 nm that is not related to SOD activity.

	1	2	3	4	5	6	7	8	9	10	11	12
A	A	A	S1	S1	S5	S5	S9	S9	S13	S13	S17	S17
B	B	B	SB1	SB1	SB5	SB5	SB9	SB9	SB13	SB13	SB17	SB17
C	C	C	S2	S2	S6	S6	S10	S10	S14	S14	S18	S18
D	D	D	SB2	SB2	SB6	SB6	SB10	SB10	SB14	SB14	SB18	SB18
E	E	E	S3	S3	S7	S7	S11	S11	S15	S15	S19	S19
F	F	F	SB3	SB3	SB7	SB7	SB11	SB11	SB15	SB15	SB19	SB19
G	G	G	S4	S4	S8	S8	S12	S12	S16	S16	S20	S20
H	BL	BL	SB4	SB4	SB8	SB8	SB12	SB12	SB16	SB16	SB20	SB20

A-G = Standard Wells
BL = Blank Wells
S1-S20 = Sample Wells
SB1-SB20 = Sample Background Wells

Figure 3. Sample plate format

Pipetting Hints

- It is recommended that a multichannel pipette be used to deliver reagents to the wells. This saves time and helps maintain more precise incubation times.
- Before pipetting each reagent, equilibrate the pipette tip in that reagent (*i.e.*, slowly fill the tip and gently expel the contents, repeat several times).
- Do not expose the pipette tip to the reagent(s) already in the well.

General Information

- The final volume of the assay is 230 µl in all the wells.
- Use the diluted assay buffers in the assay.
- It is not necessary to use all the wells on the plate at one time.
- The assay is performed at 25°C.
- All reagents, except samples and the SOD Assay Xanthine Oxidase, must be equilibrated to room temperature before beginning the assay.
- 20 samples, with sample backgrounds, can be assayed in duplicate.
- Monitor the absorbance at 450 nm using a plate reader.

Standard Preparation

Dilute 10 µl of the SOD Standard with 990 µl of Sample Buffer (1X) to obtain the SOD stock solution. Take seven clean glass test tubes and mark them A-G. Add the amount of SOD stock and Sample Buffer (1X) to each tube as described in Table 2, below.

Tube	SOD Stock (µl)	Sample Buffer (1X) (µl)	Final SOD Activity (U/ml) in Well
A	0	1,000	0
B	20	980	0.005
C	40	960	0.010
D	80	920	0.020
E	120	880	0.030
F	160	840	0.040
G	200	800	0.050

Table 2. SOD standards

Performing the Assay

If using an inhibitor, refer to the preparation of the SOD Assay Radical Detector (page 10).

1. Add the appropriate amount of reagents to the designated wells according to Table 3, below.

Reagent	SOD Standard Wells	Sample Wells	Sample Background Wells	Blank Wells	Sample Wells + Inhibitor (optional)
SOD Standard	10 µl	--	--	--	--
Sample	--	10 µl	10 µl	--	10 µl
Radical Detector Mix	100 µl	100 µl	100 µl	100 µl	--
Radical Detector + Inhibitor Mix (optional)	--	--	--	--	100 µl
Sample Buffer (1X)	--	--	120 µl	130 µl	--

Table 3. Pipetting summary

2. Initiate the reaction by adding 120 µl of the diluted SOD Assay Xanthine Oxidase to Standard, Sample and optional Sample + Inhibitor wells. Thoroughly mix by pipetting up and down, ensuring no bubbles are formed. Do not add Xanthine Oxidase to Sample Background or Blank Wells.
3. Cover the plate with the 96-Well Cover Sheet (Item No. 400012) and incubate the plate on shaker for 30 minutes at room temperature.
4. Remove plate cover. Read the absorbance at 450 nm using a plate reader.

Calculations

1. Calculate the average absorbance of each standard, sample, sample background, and blank wells.
2. Subtract the absorbance value of the blank from each standard to obtain corrected standard values (CSV).
3. Calculate the linearized rate (LR) of each standard by dividing the averaged absorbance of standard A by the averaged absorbance of each standard as shown in the equation below:

$$\text{LR of standards A to G} = \frac{CSV_{\text{std A}}}{CSV_{\text{std A to G}}}$$

4. Plot the linearized rates of each standard as a function of final SOD Activity (U/ml) from Table 2 on page 17. See Figure 4 on page 21 for a typical standard curve.
5. Calculate the linearized rate of each sample using the equation below:

$$\text{Sample LR} = \frac{CSV_{\text{std A}}}{Abs_{sample} - Abs_{sample\ background}}$$

6. Calculate the SOD activity of the samples using the equation obtained from the linear regression of the standard curve substituting the linearized rate (LR) for each sample.

$$\text{SOD (U/ml)} = \left[\left(\frac{\text{sample LR - y-intercept}}{\text{slope}}\right) \times \frac{0.23\ \text{ml}}{0.01\ \text{ml}}\right] \times \text{sample dilution}$$

One unit is defined as the amount of enzyme needed to exhibit 50% dismutation of the superoxide radical measured in change in absorbance per minute at 25°C and pH 8.0. SOD activity is standardized using the cytochrome *c* and xanthine oxidase coupled assay.

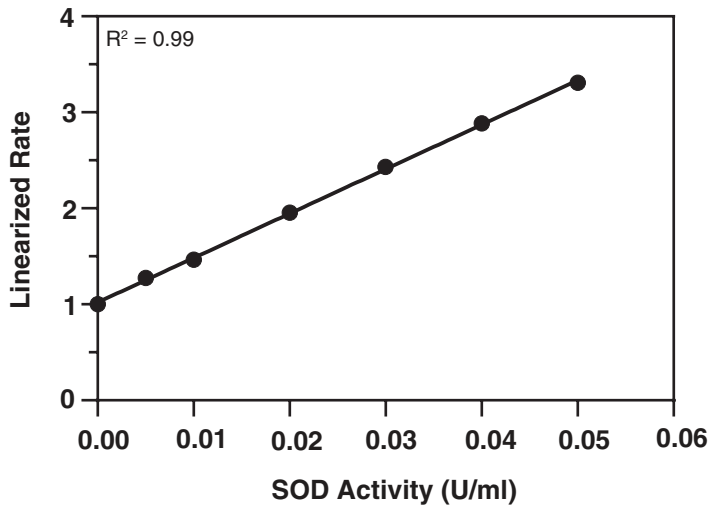


Figure 4. SOD standard curve

Performance Characteristics

Precision:

When a series of 60 SOD standard measurements were performed on the same day under the same experimental conditions, the intra-assay coefficient of variation was 3.2%. When a series of 60 SOD standard measurements were performed on five different days under the same experimental conditions, the inter-assay coefficient of variation was 3.7%.

Assay Range:

Under the standardized conditions of the assay described in this booklet, the sensitivity of the kit is 0.005 U/ml SOD.

RESOURCES

Interferences

The following reagents were tested for interference in the assay.

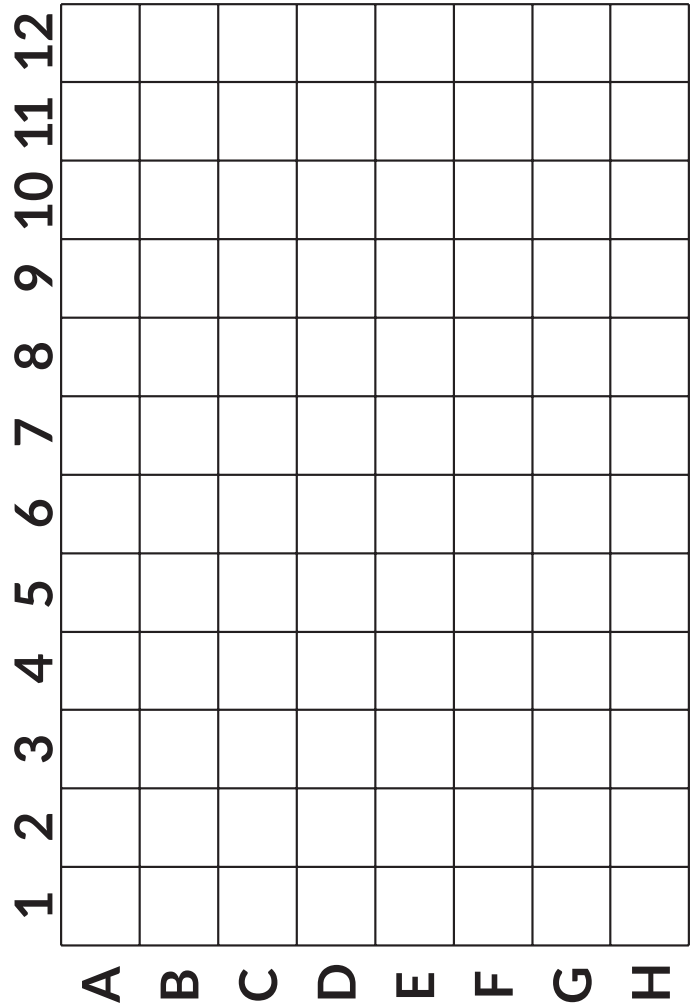
	Reagent	Will Interfere (Yes or No)
Buffers	Tris	No
	Borate	Yes
	HEPES	No
	Phosphate	No
Detergents	SDS (0.1%)	Yes
	Triton X-100 (≤0.1%)	No
	Polysorbate 20 (0.1%)	Yes
	CHAPS (0.1%)	Yes
Proteases/Inhibitors/ Chelators	Antipain (0.1 mg/ml)	Yes
	PMSF (1 mM)	Yes
	Leupeptin (1 mg/ml)	Yes
	Trypsin (0.1 mg/ml)	Yes
	Chymostatin (0.1 mg/ml)	Yes
	EGTA (≤1 mM)	No
	EDTA (≤1 mM)	No
Solvents	Ethanol (10 µl)	No
	Methanol (10 µl)	Yes
	DMSO (10 µl)	Yes
Others	Glutathione (≤1.5 mM)	No
	Glycerol (≤1%)	No
	BSA (≤1%)	No
	Dithiothreitol (3 mM)	Yes

Troubleshooting

Problem	Possible Causes	Recommended Solutions
Erratic values; dispersion of replicates	A. Poor pipetting/technique B. Bubble in the well(s)	A. Be careful not to splash the contents of the wells B. Carefully tap the side of the plate with your finger to remove bubbles
No activity was detected in the sample	SOD activity was too low or sample was too dilute	Prepare more concentrated samples
No color development in any of the wells	The radical detector was not added to the radical detector mix	Remake the radical detector mix and re-assay

References

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NOTES

Warranty and Limitation of Remedy

Buyer agrees to purchase the material subject to Cayman's Terms and Conditions. Complete Terms and Conditions including Warranty and Limitation of Liability information can be found on our website.

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